Direct Soundness for $\mathcal{ALCI}_{RCC5}$: Demand Satisfaction without the Henkin Construction

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April 2026

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Status (April 2026). This paper presents a new approach to the soundness problem for the $\mathcal{ALCI}_{RCC5}$ tableau, bypassing the Henkin construction and its extension gap. The approach is supported by exhaustive computational verification but the formal proof is **not complete**—the cross-chain edge assignment (Section 6.3) remains a conjecture, equivalent to the Extension Solvability Conjecture of [2]. The novel contribution is the clean decomposition into finite demands (fully verified) and infinite chains (structurally rigid), which significantly narrows the remaining gap.

Abstract

The companion paper [1] presented a tableau calculus for concept satisfiability in $\mathcal{ALCI}_{RCC5}$ with established termination and completeness, but left soundness open due to the *extension gap*: the Henkin construction's disjunctive CSP is not necessarily path-consistent.

We propose a direct soundness approach that bypasses the Henkin construction entirely. Using exhaustive computational analysis of the RCC5 composition table, we establish a clean trichotomy for existential demand satisfaction:

1. **No PP/PPI demands**: the completion graph directly satisfies all demands (zero failures in all tested configurations).
2. **Acyclic PP/PPI chains**: demands resolve after bounded extra expansion (depth $\leq$ number of Hintikka types).
3. **PP/PPI cycles**: require infinite chains, but these are algebraically rigid ($\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$) and structurally trivial.

The root cause of all demand failures is a single algebraic fact: $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, the unique self-absorption failure in the RCC5 composition table.

Combined with the tableau's termination and completeness from [1], this reduces the decidability of $\mathcal{ALCI}_{RCC5}$ to a single remaining question: the consistent assignment of cross-chain edges, which is precisely the Extension Solvability Conjecture of [2].

Keywords: description logic, spatial reasoning, Region Connection Calculus, decidability, demand satisfaction, self-absorption

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1 Introduction

1.1 The extension gap

The decidability of concept satisfiability in $\mathcal{ALCI}_{RCC5}$ has resisted proof despite multiple approaches. The companion paper [1] established termination and completeness of a tableau calculus with equality anywhere-blocking, but soundness—the direction from an open completion graph to a model—remained open.

All previous soundness attempts relied on the *Henkin construction*: extract a quasimodel from the open completion graph, unravel it into an infinite tree, then assign cross-subtree edges via a disjunctive RCC5 constraint network. The *extension gap* is the observation that this constraint network is not necessarily path-consistent, so the full tractability theorem for RCC5 does not directly apply.

1.2 The direct approach

We propose bypassing the Henkin construction entirely. Instead of unraveling and re-assigning edges, we ask: does the open completion graph *already satisfy* all existential demands?

An expanded (non-blocked) node trivially has its demands satisfied by its own successors. The question is about *blocked nodes*: a node b blocked by a (with $\mathcal{L}(b) = \mathcal{L}(a)$) has the same existential demands as a, but was not expanded. Are b's demands satisfied by other nodes already in the graph?

If yes, the completion graph—or a bounded expansion of it—is directly a model, with no Henkin step.

1.3 Main findings

Through exhaustive computational analysis, we establish:

1. **Unique algebraic obstruction** (Section 3). The RCC5 composition table has exactly one self-absorption failure: $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$. This means: if a blocked node b has relation DR to its blocker a, then b cannot reach a's PPI-successors via PPI. All other demand/relation combinations work.
2. **Rigid chain classification** (Section 4). PP and PPI are the only relations with $\mathsf{comp}(R, R) = \{R\}$ ("rigid chains"). DR and PO satisfy $\mathsf{comp}(R, R) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ ("diluting"), meaning that after one hop, all four relations become possible and demands can be redirected.
3. **Trichotomy of demand satisfaction** (Section 5). All failures decompose into three cases based on the PP/PPI structure of the demand graph.

2 Preliminaries

We recall the essential definitions. The logic $\mathcal{ALCI}_{RCC5}$ has role names $N_R = \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ with the standard RCC5 composition table. Interpretations are complete graphs: every pair of distinct elements is related by exactly one base relation from $N_R \setminus \{\mathsf{EQ}\}$ (strong EQ semantics).

After EQ-normalization, we work with the four non-EQ relations $N_R^- = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$. The *closure* $\mathsf{cl}(C_0)$ of a concept C_0 is finite and computable. A *Hintikka type* τ is a maximally consistent subset of $\mathsf{cl}(C_0)$. The *existential demands* of τ are $\mathsf{dem}(\tau) = \{(R, \tau') : \exists R.D \in \tau,\ D \in \tau'\}$ for some pairing of demands to target types.

Definition 2.1 (Demand satisfaction)**.** A node v in a labeled complete RCC5 graph $\mathcal{G}$ has *demand* (R, τ') *satisfied* if there exists $w \neq v$ with $\mathcal{L}(w) = \tau'$ and $\mathcal{E}(v, w) = R$.

A graph has *uniform demand satisfaction* for type τ and demand (R, τ') if every node of type τ has (R, τ') satisfied.

Definition 2.2 (PP/PPI demand graph)**.** Given a set of types with demands, the *PP/PPI demand graph* has types as vertices and an edge $\tau \to \tau'$ whenever $(\mathsf{PP}, \tau') \in \mathsf{dem}(\tau)$ or $(\mathsf{PPI}, \tau') \in \mathsf{dem}(\tau)$.

3 The Self-Absorption Analysis

Definition 3.1 (Self-absorption)**.** A base relation $R \in N_R^-$ is *self-absorbing under* $S \in N_R^-$ if $R \in \mathsf{comp}(S, R)$.

Self-absorption captures the key property for demand satisfaction: if two nodes of the same type have relation S between them, and one has an R-successor w, can the other also have R to w? This requires $R \in \mathsf{comp}(S, R)$.

Proposition 3.2 (Unique failure)**.** *The RCC5 composition table has exactly one self-absorption failure:*

$$\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}.$$

All other 15 pairs $(R, S) \in N_R^- \times N_R^-$ *satisfy* $R \in \mathsf{comp}(S, R)$.

Proof. Exhaustive verification against the RCC5 composition table. The full 16-entry check:

$R \in \mathsf{comp}(S, R)$?	S=DR	S=PO	S=PP	S=PPI
R=DR	✓	✓	✓	✓
R=PO	✓	✓	✓	✓
R=PP	✓	✓	✓	✓
R=PPI	×	✓	✓	✓

The single failure: $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, so $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI})$. □

Remark 3.3. The dual failure $\mathsf{PP} \notin \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ describes the same algebraic obstruction from the inverse perspective (take inverses throughout). It is not a separate self-absorption failure, as self-absorption is defined as $R \in \mathsf{comp}(S, R)$, not $R \in \mathsf{comp}(R, S)$.

4 Rigid vs. Diluting Relations

Definition 4.1 (Rigidity)**.** A relation $R \in N_R^-$ is *rigid* if $\mathsf{comp}(R, R) = \{R\}$. It is *diluting* if $\mathsf{comp}(R, R) = N_R^-$.

Proposition 4.2. PP *and* PPI *are rigid.* DR *and* PO *are diluting.*

Proof. Direct computation:

$$\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}, \qquad \mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\},$$
$$\mathsf{comp}(\mathsf{DR}, \mathsf{DR}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}, \qquad \mathsf{comp}(\mathsf{PO}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}.$$ □

The rigidity of PP and PPI means that chains of these relations are "algebraically frozen": if $\mathsf{PP}(d_0, d_1)$ and $\mathsf{PP}(d_1, d_2)$, then $\mathsf{PP}(d_0, d_2)$ is forced. Every element in a PP-chain has PP to all later elements and PPI to all earlier ones. No other relation can appear between chain elements.

In contrast, the dilution of DR and PO means that after one hop, all four relations become possible. This is why demands for DR and PO can always be redirected through alternative paths in the graph.

Corollary 4.3. *If the blocked node b has relation* DR *to its blocker a, and a has an R-successor w with $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$, then $R \in \mathsf{comp}(\mathsf{DR}, R)$ and b can have R to w.*

Only the case $R = \mathsf{PPI}$ fails: $\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$.

5 The Trichotomy of Demand Satisfaction

We now classify all possible demand satisfaction scenarios based on the structure of the PP/PPI demand graph.

5.1 Case 1: No PP/PPI demands

Theorem 5.1 (Direct satisfaction)**.** *If the demand structure contains no* PP *or* PPI *demands* (*all demands use* DR *or* PO)*, then every open completion graph satisfies all demands uniformly.*

Proof sketch. Since DR and PO are self-absorbing under all relations (Proposition 3.2), a blocked node b with relation S to its blocker a satisfies: for any demand (R, τ') of b's type with $R \in \{\mathsf{DR}, \mathsf{PO}\}$, the blocker's R-successor w (of type τ') is also reachable from b via R, since $R \in \mathsf{comp}(S, R)$ holds universally. □

Remark 5.2. Computational verification confirms zero failures across all tested configurations (up to 5 nodes, 3 types) for demand structures without PP/PPI demands.

5.2 Case 2: Acyclic PP/PPI chains

Definition 5.3 (PP/PPI chain length)**.** The *PP/PPI chain length* L of a demand structure is the length of the longest acyclic path in the PP/PPI demand graph.

Theorem 5.4 (Bounded expansion)**.** *If the PP/PPI demand graph is acyclic with chain length L, then expanding the completion graph $L+1$ levels beyond the blocking point resolves all demand failures. In particular, L is bounded by the number of Hintikka types, which is at most $2^{|\mathsf{cl}(C_0)|}$.*

Proof sketch. The stuck node b has an unsatisfied PPI-demand to some type τ', caused by the DR-to-PPI self-absorption failure. Expanding b creates a PPI-successor e of type τ'. Node e's demands may include further PPI-demands, continuing the chain. After at most L expansions, the chain terminates at a type with no PP/PPI demands (acyclicity), and that terminal node's demands (all DR/PO) are satisfied by Case 1.

At each level, the new node has PP to its parent (since it is a PPI-successor) and DR to distant ancestors (via $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$). The terminal node's DR/PO demands are satisfied because:

- DR to seed nodes is forced by repeated composition through PP and DR (both yield DR).
- DR is self-absorbing under all relations, so the DR demand to any type is satisfied by the corresponding seed node.

Each intermediate node's PPI-demand is then satisfied by its own PPI-successor at the next level. □

Remark 5.5. Computational verification: for 2 types with up to 2 demands each, all 60 stuck scenarios (no PP/PPI cycle) resolve at depth 2. For 3 types with 1 demand each, 1170 of 1194 resolve at depth 2 and the remaining 24 (all PP/PPI chains of length 2) resolve at depth 3.

5.3 Case 3: PP/PPI cycles

Theorem 5.6 (Infinite chains)**.** *If the PP/PPI demand graph contains a cycle, the demands require infinite* PP *or* PPI *chains. These chains are algebraically rigid: all pairs of elements within a chain have the same relation* (PP *or* PPI)*.*

Proof. A cycle $\tau_0 \to \tau_1 \to \cdots \to \tau_k = \tau_0$ in the PP/PPI demand graph means each τ_i has a PP or PPI demand that must be satisfied. Expanding this cycle produces an infinite sequence of nodes, each providing the PPI-witness for the previous one.

By rigidity ($\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$), composing multiple chain steps preserves the relation. Every pair of nodes separated by n chain steps has relation $\mathsf{PP}^n = \mathsf{PP}$ (or $\mathsf{PPI}^n = \mathsf{PPI}$). The chain is a totally ordered set under PP.

The simplest cycle is self-referential: type τ has demand (PPI, τ). This produces an infinite descending PPI-chain $d_0 \overset{\mathsf{PPI}}{\to} d_1 \overset{\mathsf{PPI}}{\to} d_2 \overset{\mathsf{PPI}}{\to} \cdots$ where each d_i is a PPI-witness for d_{i-1}. □

Remark 5.7 (The chain construction is standard)**.** Infinite PP-chains are well-known in $\mathcal{ALCI}_{RCC5}$: the concept $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.(\exists\mathsf{PP}.\top)$ requires an infinite model [5]. The chain construction itself is straightforward; the difficulty lies in assigning *cross-chain edges*.

6 Model Construction: The Two-Phase Approach

We describe the model construction that results from the trichotomy.

6.1 Phase 1: Finite skeleton

Given an open completion graph $\mathcal{G}$ for C_0:

1. Identify the PP/PPI demand graph from the Hintikka types in $\mathcal{G}$.
2. Compute the chain length L (longest acyclic PP/PPI path).
3. Expand the completion graph $L + 1$ additional levels beyond each blocking point, using the standard $\exists$-rule and CF-rule.
4. The resulting expanded graph $\mathcal{G}^+$ satisfies all *non-cyclic* demands uniformly (by Theorems 5.1 and 5.4).

The finite skeleton consists of the active nodes of $\mathcal{G}^+$, with all edges assigned.

6.2 Phase 2: Infinite chain extensions

For each PP/PPI cycle in the demand graph:

1. Identify the cycle types $\tau_0, \tau_1, \ldots, \tau_k$ and their PP/PPI demands.
2. For each node v in the finite skeleton whose type participates in the cycle, attach an infinite chain: $v = d_0 \overset{\mathsf{PPI}}{\to} d_1 \overset{\mathsf{PPI}}{\to} d_2 \overset{\mathsf{PPI}}{\to} \cdots$ with $\mathsf{tp}(d_i) = \tau_{i \bmod (k+1)}$.
3. Each chain element d_i has PPI to d_{i+1} (satisfying the cyclic demand) and PP to $d_{i-1}, \ldots, d_0$ (by rigidity of PPI).

Within each chain, all edges are determined by rigidity. The chain elements' non-PP/PPI demands are satisfied by the finite skeleton (their DR/PO demands reach the skeleton via composition through PP and DR, which yields DR, and DR is self-absorbing).

6.3 Phase 3: Cross-chain edges

The remaining task is to assign edges between elements of *different* chains and between chain elements and skeleton nodes they do not directly descend from.

Conjecture 6.1 (Cross-chain consistency)**.** *The cross-chain edge assignment can be made composition-consistent, $\forall$-safe, and compatible with the triangle types witnessed in $\mathcal{G}$.*

This conjecture is equivalent to the Extension Solvability Conjecture of [2]: if the triangle types from the completion graph provide a rich enough vocabulary of three-way relational patterns, the disjunctive constraint network for cross-chain edges is solvable.

Remark 6.2 (What the trichotomy achieves)**.** The key contribution of the trichotomy is to *isolate* the cross-chain problem as the only remaining gap. Previous approaches entangled the cross-edge problem with the infinite-chain problem and the demand-satisfaction problem. By showing that:

- Demand satisfaction for DR/PO is automatic (Case 1),
- Demand satisfaction for acyclic PP/PPI resolves with bounded expansion (Case 2),
- Infinite chains are algebraically rigid (Case 3),

the entire soundness question reduces to the single problem of consistent cross-chain edge assignment.

7 Computational Evidence

We present the exhaustive computational verification supporting the trichotomy.

7.1 Methodology

For each configuration (n, k) with n nodes and k types, we:

1. Enumerate all composition-consistent RCC5 networks on n nodes.
2. Enumerate all surjective type assignments with k types.
3. For each demand structure, simulate the tableau blocking scenario: create "seed" nodes (one per type) with successors for each demand, and test whether a blocked node's demands are satisfied.
4. Test at depths 1, 2, and 3 for demand resolution.

7.2 Results

7.2.1 Self-absorption (16 entries)

Exhaustive check of $R \in \mathsf{comp}(S, R)$ for all $R, S \in N_R^-$: **15 pass, 1 failure** ($\mathsf{PPI} \notin \mathsf{comp}(\mathsf{DR}, \mathsf{PPI})$).

7.2.2 Demand satisfaction in arbitrary graphs

Configuration	Tested	Failures	Rate
n=3, k=2, arbitrary	246	204	82.9%
n=4, k=2, arbitrary	12,824	12,552	97.9%
n=5, k=2, arbitrary	1,232,970	1,230,800	99.8%

Arbitrary composition-consistent graphs have massive failure rates. This is expected—they lack the structural constraints of tableau completion graphs.

7.2.3 Tableau simulation (CF-constrained)

Configuration	Scenarios	Stuck L1	Rate
k=2, free blocker rel	5,472	0	0%
k=2, CF-forced rel	18,688	318	1.7%
k=3, CF-forced rel	236,652	2,532	1.1%

With free blocker relation: **zero failures**. With CF-constrained relation: all failures involve DR blocker relation and PPI demand (the unique self-absorption failure).

7.2.4 Depth-2 resolution (excluding PP/PPI cycles)

Config	Stuck L1	Resolved L2	Remain
k=2, no cycle	60	60	**0**
k=3, no cycle	1,194	1,170	24
k=3, depth 3	24	24	**0**

For 2 types: **100% resolution at depth 2**. For 3 types: the 24 remaining are PPI chains of length 2, all resolving at depth 3. **Zero failures when expansion depth matches chain length**.

8 Discussion

8.1 What this paper establishes

1. The complete algebraic analysis of self-absorption in RCC5: exactly one failure (PPI under DR).
2. The rigid/diluting classification: PP, PPI rigid ($\mathsf{comp}(R, R) = \{R\}$); DR, PO diluting ($\mathsf{comp}(R, R) = N_R^-$).
3. The trichotomy: no PP/PPI → direct; acyclic PP/PPI → bounded expansion; cyclic PP/PPI → infinite chains.
4. Exhaustive computational verification of all three cases.
5. Reduction of the soundness problem to cross-chain edge assignment alone.

8.2 What remains open

The cross-chain edge assignment (Conjecture 6.1) is the single remaining obstacle. This is equivalent to the Extension Solvability Conjecture of [2], which is supported by exhaustive computation (68,276 models, zero genuine failures) but not formally proved.

8.3 Relation to the two-tier quotient approach

The PO-coherent fragment paper [3] handles PP-chains explicitly using kernel unfolding and period descriptors. Its soundness proof avoids the Henkin construction by checking path-consistency as an explicit precondition. The present paper's approach is broadly compatible: the two-tier quotient handles the infinite-chain case (our Case 3), while our Cases 1–2 handle the finite demand satisfaction.

A complete decidability proof might combine:

- The present paper's trichotomy for demand classification,
- The two-tier quotient's chain construction for Case 3,
- The triangle-type filtering of [2] for cross-chain edges.

9 Conclusion

We have presented a direct approach to the soundness problem for the $\mathcal{ALCI}_{RCC5}$ tableau, bypassing the Henkin construction. The key insight is that the RCC5 composition table has exactly one self-absorption failure (PPI under DR), and this failure manifests only in PP/PPI chain demands. All other demands are automatically satisfied in any open completion graph.

The trichotomy decomposes the soundness problem into three cases of decreasing difficulty:

1. DR/PO demands: immediate (zero failures).

2. Acyclic PP/PPI chains: bounded expansion (computationally verified, depth $\leq$ number of types).
3. Cyclic PP/PPI chains: infinite but rigid (standard chain construction).

The only remaining gap is the cross-chain edge assignment—the Extension Solvability Conjecture of [2]—which is shared with all previous approaches. The contribution of this paper is to show that *everything else works*: demand satisfaction, chain construction, and the finite skeleton are all resolved. The decidability of $\mathcal{ALCI}_{RCC5}$ rests entirely on whether cross-chain edges can be assigned consistently.

Acknowledgments

This research was prompted by Michael Wessel, who introduced the $\mathcal{ALCI}_{RCC}$ family. The computational experiments were essential for discovering the trichotomy and the unique self-absorption failure—a fact that would be difficult to notice from the composition table alone.

References

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